P.E.S. COLLEGE OF ENGINEERING, MANDYA, 571401

(An Autonomous Institution under Visvesvaraya Technological University, Belagavi)

Assignment on,

COMPUTER NETWORKS (P22CS502)

Course Instructor: - **Namratha mam**

Assistant professor of computer science and engineering

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
P.E.S. College of Engineering, Mandya.

AY 2025-26

SEM: 5th **SEC**: A

NAME	**USN**	**MARKS AWARDED**
Hrushith	4PS23CS059	
Kannika k m	4PS23CS064	
Karukonda Tirumala	4PS23CS067	
Kumar Kislay	4PS23CS074	
Hamsa Priya	4PS23CS191	
Sinchana	4PS23CS192	
Rakshitha	4PS23CS196	

5G-Advanced (5G-A) and Network Slicing

1.INTRODUCTION

5G-Advanced (5G-A) is the next major evolution of 5G networks, introduced in 3GPP Release 18. It enhances the performance, intelligence, reliability, and efficiency of existing 5G systems. 5G-A bridges the gap between 5G and 6G, enabling more advanced applications like autonomous systems, immersive AR/VR, smart manufacturing, and large-scale IoT.

A key enabler of 5G-A is Network Slicing — a technology that allows operators to create multiple virtual networks over a single physical infrastructure. Each slice is customized for a specific type of service (e.g., IoT, emergency services, high-speed mobile broadband).

5G-A combined with network slicing delivers:

- Ultra-low latency
- Higher throughput
- Enhanced reliability
- Energy efficiency
- AI-driven network optimization

2. WHAT IS 5G-ADVANCED (5G-A)?

5G-Advanced is an enhanced version of 5G that introduces new capabilities beyond standard 5G networks.

Key Features of 5G-A:

- AI-native networking – AI is used for traffic prediction, interference management, and fault detection.
- Enhanced Mobile Broadband (eMBB++) – Higher peak speeds and improved coverage.
- Improved URLLC – Ultra-reliable low-latency communication for mission-critical applications.
- Massive IoT improvements – Better efficiency and battery life for IoT devices.

• Energy-efficient network operations – Smart power saving and green communication.
• Support for XR (Extended Reality) – Smooth AR/VR/MR experiences.

3. WHAT IS NETWORK SLICING?

Network slicing creates independent, end-to-end virtual networks on a shared 5G infrastructure.

Each slice has:

✓ Dedicated bandwidth
✓ Customized latency
✓ Individual security rules
✓ Independent QoS policies

Types of Slices:

1. eMBB Slice – High-speed internet for HD video, gaming, conferencing.

2. URLLC Slice – Extremely low latency for autonomous vehicles, remote surgery.

3. mMTC Slice – Connects millions of IoT devices with low power consumption.

With 5G-A, slicing becomes more dynamic, automated, and AI-driven.

4. COMPONENTS OF 5G-A & NETWORK SLICING

4.1 Radio Access Network (RAN)

- 5G-A uses massive MIMO, beamforming, and AI-optimized RAN.
- Supports multiple slices at the RAN level.

4.2 Core Network

The 5G core allows full customization of:
- Latency
- Bandwidth
- Security
- Mobility handling

Functions like AMF, SMF, UPF can be customized per slice.

4.3 Orchestration & Automation

5G-A introduces:
- AI/ML based slice management
- Real-time scaling of resources
- Self-healing networks

Tools:
- NFV (Network Function Virtualization)
- SDN (Software Defined Networking)

---

5. ARCHITECTURE OF 5G-A NETWORK SLICING

The architecture is typically end-to-end and includes:

5.1 Service Layer

- Defines services like IoT slice, emergency slice, broadband slice.
- Provides APIs for applications to request slices.

5.2 Network Slice Management Layer

Handles:

- Slice creation
- Provisioning
- Monitoring
- Scaling

Uses AI to optimize traffic and resources.

5.3 Infrastructure Layer

- Physical towers, fiber, servers, edge computing.
- Virtualized to support multiple slices simultaneously.

---

6. NEED FOR 5G-A AND NETWORK SLICING

1. Support Diverse Applications
Gaming, IoT, autonomous cars, healthcare all need different performance levels.

2. Efficient Network Utilization
Slices help allocate resources dynamically without wasting bandwidth.

3. Mission-Critical Services
URLLC slice ensures 99.999% reliability.

4. Business Customization
Enterprises can get private slices for industries like manufacturing, mining, ports.

5. Scalable IoT Deployment
Millions of devices can connect with low power usage.

6. Better Quality of Experience (QoE)
Users get faster and more stable connectivity.

---

7. REAL-WORLD USE CASES OF 5G-A WITH NETWORK SLICING

7.1 Smart Manufacturing

- Real-time robotic control
- Predictive maintenance
- Autonomous guided vehicles

7.2 Autonomous Vehicles

- Ultra-low latency slice
- Vehicle-to-everything (V2X) communication

7.3 Smart Cities

- Public safety slice
- IoT slice for sensors
- High-speed slice for citizens

7.4 Healthcare

- Remote surgery

- Telemedicine
- Wearable monitoring devices

7.5 Entertainment & XR

- AR/VR gaming
- Metaverse platforms

---

8. SECURITY IN 5G-A & NETWORK SLICING

Major Risks:

- Slice isolation failures
- Core network attacks
- IoT vulnerabilities
- DDoS attacks

Security Measures:

- Strong isolation between slices
- AI-based threat detection
- End-to-end encryption
- Zero-trust architecture
- Continuous monitoring

---

9. CHALLENGES IN 5G-A AND NETWORK SLICING

1. High deployment cost

2. Complex slice management

3. Interoperability between vendors

4. Need for skilled workforce

5. Ensuring slice isolation and security

6. Regulatory issues and spectrum allocation

---

10. FUTURE SCOPE

- Evolution towards 6G Intelligent Networks
- Fully autonomous networks (Self-X capabilities)
- Advanced AR/VR/XR services
- Network slicing for space–air–ground integrated networks
- AI-native end-to-end optimization
- Massive digital twin applications

---

11. CONCLUSION

5G-Advanced and Network Slicing together form the foundation of next-generation communication systems. 5G-A brings improved speed, reliability, intelligence, and energy efficiency. Network slicing allows service providers to deliver multiple customized virtual networks over a single infrastructure.

This combination supports future applications such as autonomous vehicles, smart cities, immersive XR experiences, and mission-critical services. As technology evolves toward 6G, 5G-A will play a key role in creating intelligent, flexible, and efficient network ecosystems.